

tester that generates up to 1 kV."

Harmonic ratio measurement (for maintenance of facilities, generators and transformers), switch count (for detection of glitches in switch and relay systems) and built-in square wave generation (for designing and troubleshooting motor drivers) are some of the enhanced measurement capabilities offered by companies today.

Bob Green, senior market development manager, Keithley, says, "More intelligence has been incorporated in some digital multimeters (DMMs) so that these can act as partial controllers. Code can be downloaded into the instrument so that the instrument can control other instruments and make decisions. It is designed to reduce bus traffic and speed-up test processing. Some DMMs include switching capability for automated testing. More intelligence has been placed on switch cards to monitor relay life with switch closure counts and to automatically identify the card type and connect it to an analogue backplane."

Data logging capabilities

Multimeters today can store the readings or perform automated data logging. It's an essential feature when repetitive test is required for qualification or troubleshooting.

"Parameters like short-term stability, drift and fluctuations are critical to understand a circuit's behaviour in either controlled or uncontrolled environments," explains Manjunath. He adds, "In some cases, to troubleshoot a device, the engineer may find himself looking for elusive problems like intermittent dropouts or glitches. When tasked to measure these parameters, he may have to gather multiple instruments and write custom software routines to capture and analyse thousands of measurement data points."

"A common test would be voltage measurements at room, hot and cold temperatures. In such cases, it's more convenient and efficient to automate recording of measurements while you perform other tasks at hand. It's even



Dynalox DX-6101
handheld DMM

better if you can record as many data points as required to further analyse the situation without worrying about the storage capacity of the instrument," says Siddique.

Justifying with an example, Vineet Aggarwal, product manager, National Instruments, says, "In the semiconductor industry, for chip testing there are a number of different parameters to be measured. Without automation, it would be very difficult to perform complex tests. Having the software that can run through test sequences, integrating the channels and performing necessary switching operations are key aspects of PC-based ATE systems." In addition, some modular instruments have built-in synchronisation for stimulus-response testing.

"Many vendors have invested in providing an efficient software for data logging. These software are trend-plot-style graphical representation of a logged session which has been recorded by the meter. Today, not only can the user immediately view recorded results from a logging session which could have been up to 200 hours long, allowing for an instant analysis of any unusual event without the need of a PC, multiple recording sessions can also be completed and viewed right on the meter without the

need for a download first. In addition to being able to view the trend information right on the screen, a cursor is available to highlight a specific event (including event detail). You can also zoom in for a closer look at the data," explains Thakkar.

Advancement in ADCs

Analogue-to-digital converters (ADCs) are at the heart of multimeters. So when talking of multimeters, it is important to mention the improvements in ADCs in terms of sampling rate, accuracy, form-factor as well as price.

Aggarwal says, "While DMMs and oscilloscopes are distinctly different equipment, the advancement in ADCs and other analogue components has created a lot of flexible designs."

"While selecting an instrument, typically, the trade-off that you make is faster sampling rate and less precision, or slower sampling rate and more precision. But with advanced ADC technology, you can get both faster sampling and high precision."

Technologies like FlexADC used in high-end digitisers and multimeters give you flexible precision and dynamic range for dynamic measurements. You can switch the sample rate as per your measurement need.

Aggarwal comments, "Earlier, in the DMM architecture, a combination of an ADC and a multiplexer was used to reduce the cost, ADC being the most expensive component of the system. But now because of the overall ADC technology, the prices have come down. We are looking at devices which have more than one ADC." He explains, "In a switching system, if you look at mul-

Reading the digits of multimeter

Rakesh Joshi, manager, Dynalog India, shares, "By convention, a half digit can display either a zero or a one, while a three-quarters digit can display a numeral higher than a one but not nine. Commonly, a three-quarters digit refers to a maximum value of 3 or 5. The fractional digit is always the most significant digit in the displayed value. A 3½-digit meter can display a quantity from 0 to 3999 or 5999, depending on the manufacturer. While a digital display can easily be extended in precision, the extra digits are of no value if not accompanied by care in the design and calibration of analogue portions of the multimeter. Meaningful high-resolution measurements require a good understanding of the instrument specifications, good control of the measurement conditions, and traceability of the calibration of the instrument."